

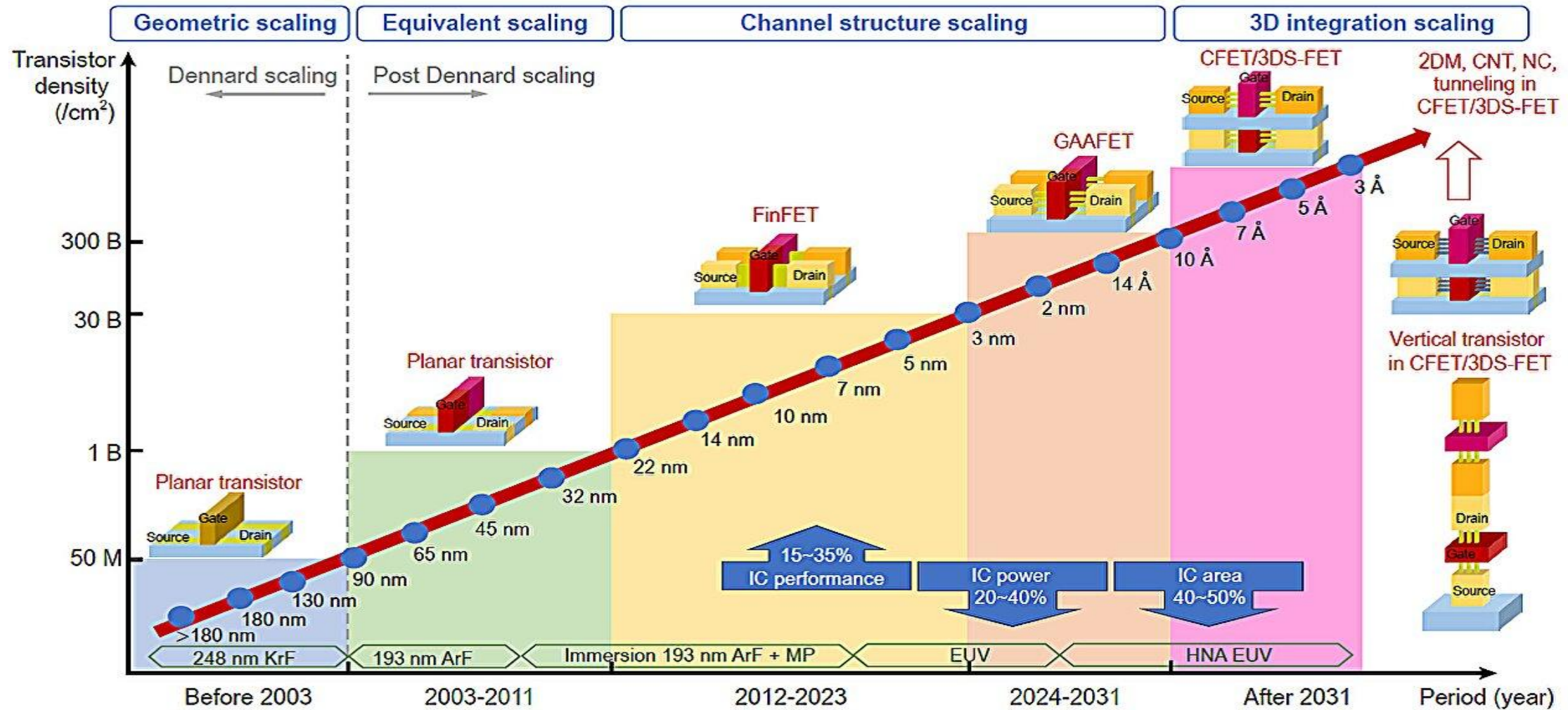
Machine Learning Driven Device Co-Optimization

-PARTH DEMBLA

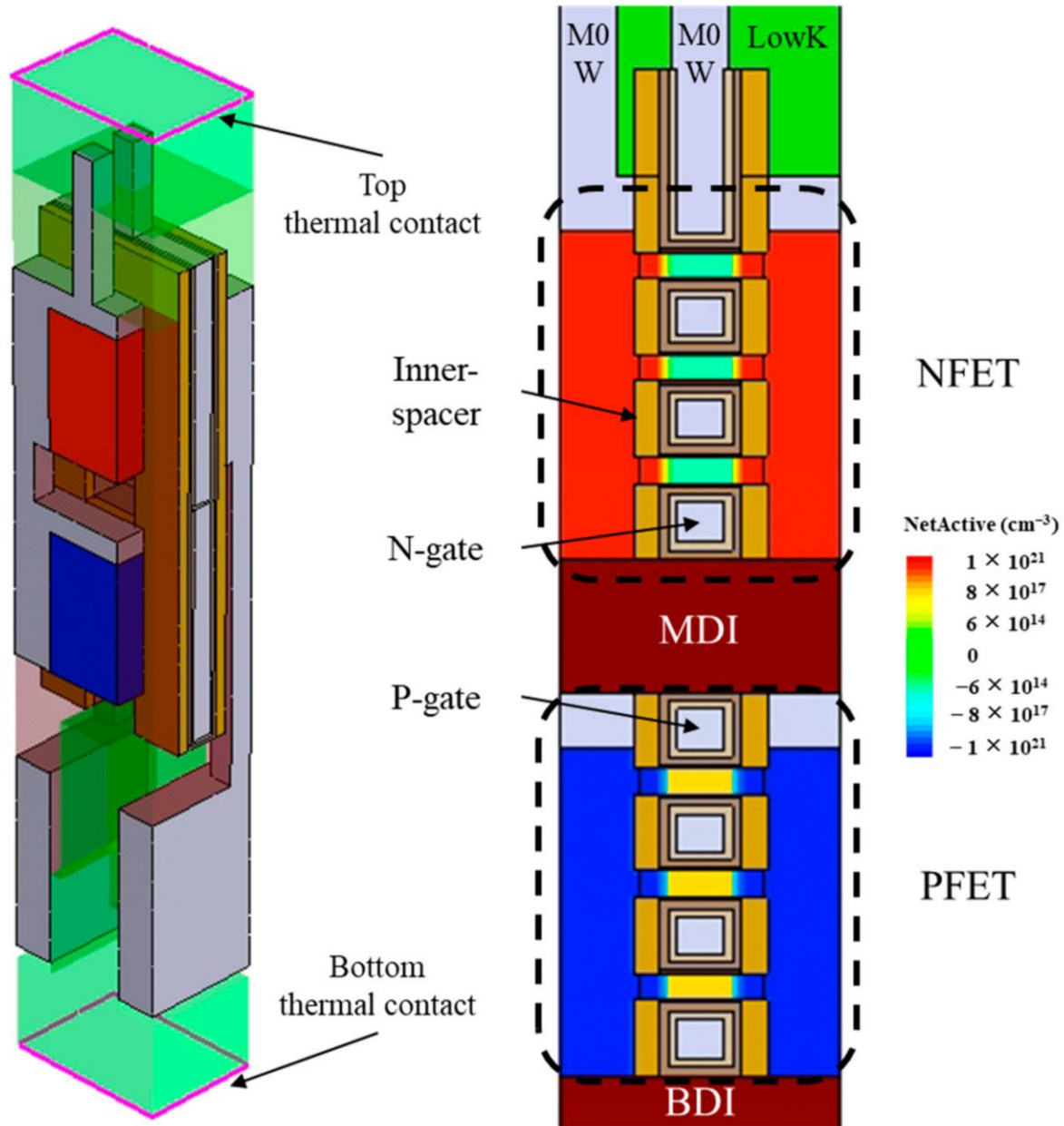
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CMOS Scaling :-



*source : New structure transistors for advanced technology node CMOS ICs



- Complementary FET structure
- Successor to GAAFET
- A 3D stacked device architecture with NFET over PFET
- Higher transistor density (no side-by-side layout)
- Saves Chip Area

- As transistor become smaller, structures become more complex, circuits become more highly integrated, and optimization of nonlinear device performance becomes more difficult

One of the key trade – off in IC design => Speed(Delay) + Efficiency (Power)

Speed => increase with increase in I_{ON}
 Static Power Consumption => decrease with decrease in I_{off} (less leakage)
 Dynamic Power Consumption => related with switching

$$P = \underbrace{\frac{1}{2}\alpha C V_{DD}^2 f}_{\text{Dynamic Power}} + \underbrace{I_{leak} V_{DD}}_{\text{Static Power}}$$

α : activity factor C : capacitance
 I_{leak} : leakage current V_{DD} : supply voltage
 f : frequency

Why optimization is important :-

- Reducing **delay** makes circuits faster.
- Reducing power improves **thermal reliability** and **reduces packaging cost**.

Logic ICs are used in various devices and appliances.

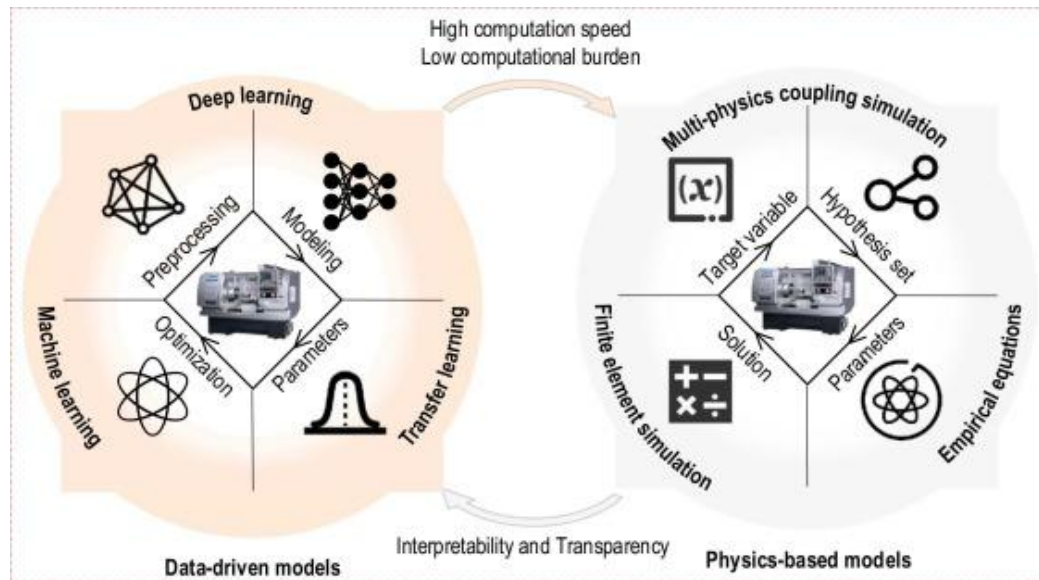


Problem Statement:-

Predict and optimize the inverter design {width(W), length(L), number of sheets(N)} to meet specified power and delay requirements for advanced nanosheet FET and CFET-based technologies

Solution Approach

1. Developing an analytical model for inverter delay and power as functions of key parameters such as effective current and capacitance.
 2. Build a Deep Neural Network (DNN)-based forward model to predict current and capacitance values for inverter design parameters (W, L, N). Implementing optimization algorithms like Bayesian or grid search .
-
- TCAD simulations of state-of-the-art devices are time-consuming.
 - These simulations incur high computational costs to obtain optimal design parameters before Manufacturing Process Evaluation ; that's where ML as a solution can work



Why to develop an analytical model?

DATA DRIVEN MODEL

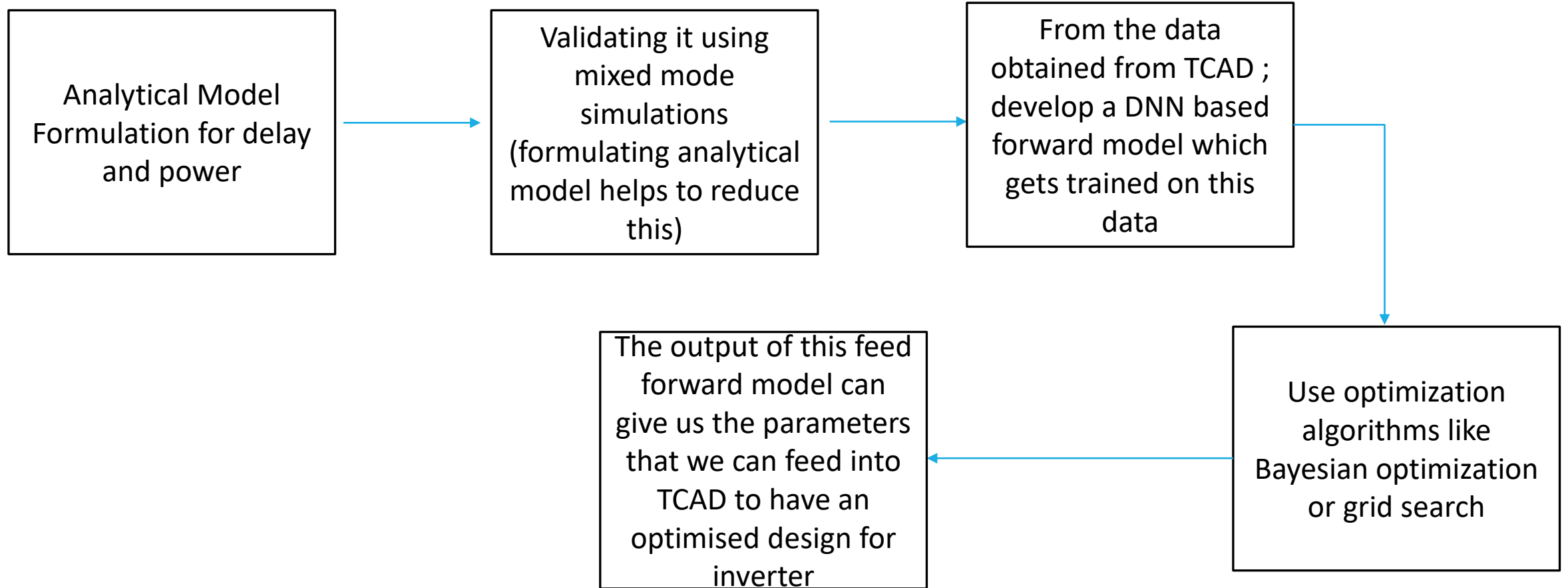
1. Use machine learning (ML), deep learning (DL), and statistical methods to model a system by learning from experimental or simulated data
2. Cleaning and preparing data for modeling; fine tuning ; extraction of data without physical interpretability
3. May fail in unseen scenarios not seen in training
4. hence difficult to generalise

PHYSICS BASED MODEL

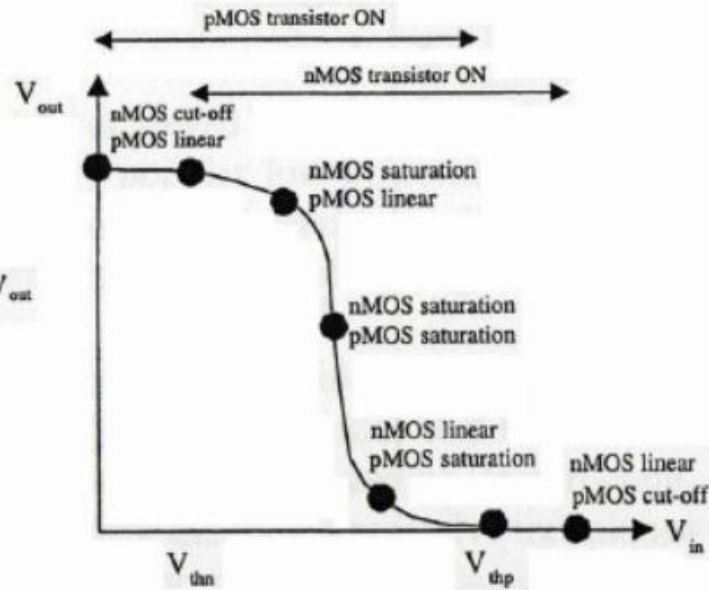
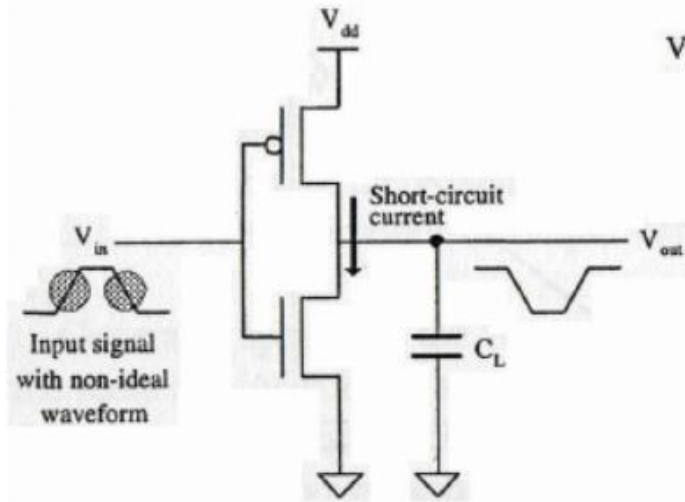
1. Use mathematical equations derived from physical principles
2. High interpretability and transparency
3. Generalizable and robust to new conditions.
4. Time-consuming to simulate especially in complex, multi-scale systems.

*If from an analytical model , we know what are the current and capacitance ranges to look for , we can proceed accordingly

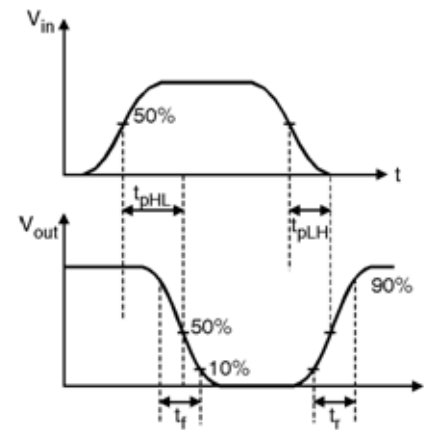
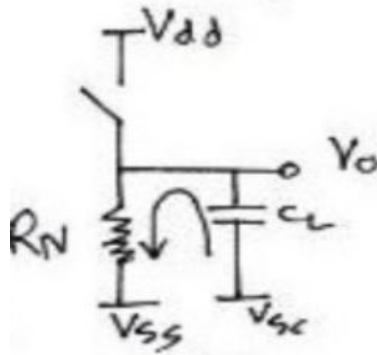
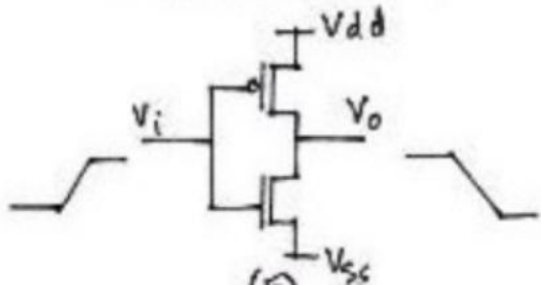
METHODOLOGY



Basics of inverter



*source : K. Eshraghian, D.A. Pucknell, and S.M. Kang, *CMOS Digital Integrated Circuits: Analysis and Design*, 3rd ed. New York: McGraw-Hill, 2003.



Propagation delay and rise and fall times

Existing Delay Models Used

$$t_{pd} = \int_{V_{initial}}^{V_{final}} \frac{C_L dV}{I_{eff}}$$

$I_{effective}$ => To model

1) One Point Model

- Uses the equation such that $I_{effective}$ is taken as I_{ON}
- I_{ON} was taken as $I_{dsat} = I_d(V_{GS}=V_{DD}, V_{DS} = V_{DD})$

Why I_{ON} was used? :-

- It represents the peak drive strength of the transistor.
- The actual transient current does not remain constant but stays close to this value during a significant portion of switching

One Point Model:-

$$\tau_{PHL} = C_L V_{DD} / 2I_{ON(NMOS)}$$

$$\tau_{PLH} = C_L V_{DD} / 2I_{ON(PMOS)}$$

$$\tau_P = (\tau_{PLH} + \tau_{PHL})/2$$

For **falling delay** (t_{pHL}):

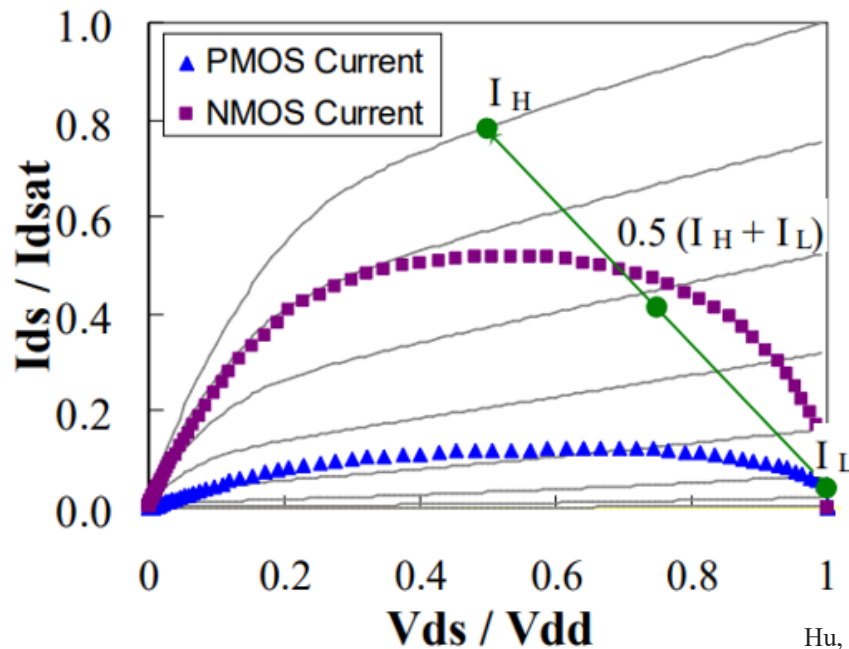
Time taken for the output to fall from **high** to **50% of V_{DD}** after the input crosses 50% (rising).

For **rising delay** (t_{pLH}):

Time taken for the output to rise from **low** to **50% of V_{DD}** after the input falls through 50%.

2) Two Point Model For Effective Drive Current

- The actual delay of a CMOS inverter depends on the transient current profiles during switching, which are influenced by the varying V_{GS} & V_{DS} voltages.
- I_{dsat} , being constant, does not account for these variations and fails to capture the dynamics of switching accurately; hence one point model is not that accurate (under estimation of delay)
- New model proposed which could capture transient behavior as well



Two Point Model:-

$$I_{\text{effective(NMOS)}} = (I_{NL} + I_{NH}) / 2$$

$$I_{\text{effective(PMOS)}} = (I_{PL} + I_{PH}) / 2$$

$$\tau_{PHL} = C_L V_{DD} / 2I_{\text{effective(NMOS)}}$$

$$\tau_{PLH} = C_L V_{DD} / 2I_{\text{effective(PMOS)}}$$

$$\tau_P = (\tau_{PLH} + \tau_{PHL}) / 2$$

$$I_H = I_{ds} (V_{GS} = V_{dd} \text{ and } V_{DS} = 0.5V_{dd})$$

$$I_L = I_{ds} (V_{GS} = 0.5 V_{dd} \text{ and } V_{DS} = V_{dd})$$

Hu, Jenny, et al. "Effective drive current in CMOS inverters for sub-45 nm technologies." *Proc. NSTI Bio-NanoTech. Conf. Trade Show*. Vol. 3. 2008.

3) Three Point Model

- To introduce another transient point in between ; a three point model approach is proposed

Three Point Model:-

$$I_{\text{effective(NMOS)}} = (I_{\text{NL}} + I_{\text{NH}} + I_{\text{NM}}) / 3$$

$$I_{\text{effective(PMOS)}} = (I_{\text{PL}} + I_{\text{PH}} + I_{\text{PM}}) / 3$$

$$\tau_{\text{PHL}} = C_L V_{\text{DD}} / 2I_{\text{effective(NMOS)}}$$

$$\tau_{\text{PLH}} = C_L V_{\text{DD}} / 2I_{\text{effective(PMOS)}}$$

$$\tau_P = (\tau_{\text{PLH}} + \tau_{\text{PHL}}) / 2$$

$$I_H = I_{\text{ds}} (V_{\text{GS}} = V_{\text{DD}} \text{ and } V_{\text{DS}} = 0.5V_{\text{DD}})$$

$$I_L = I_{\text{ds}} (V_{\text{GS}} = 0.5 V_{\text{DD}} \text{ and } V_{\text{DS}} = V_{\text{dd}})$$

$$I_M = I_{\text{ds}}(V_{\text{GS}}=0.75 V_{\text{DD}}, V_{\text{DS}}=0.75 V_{\text{DD}}).$$

To start formulating the analytical model , inverter was simulated in cadence simulation software by taking different load capacitances.

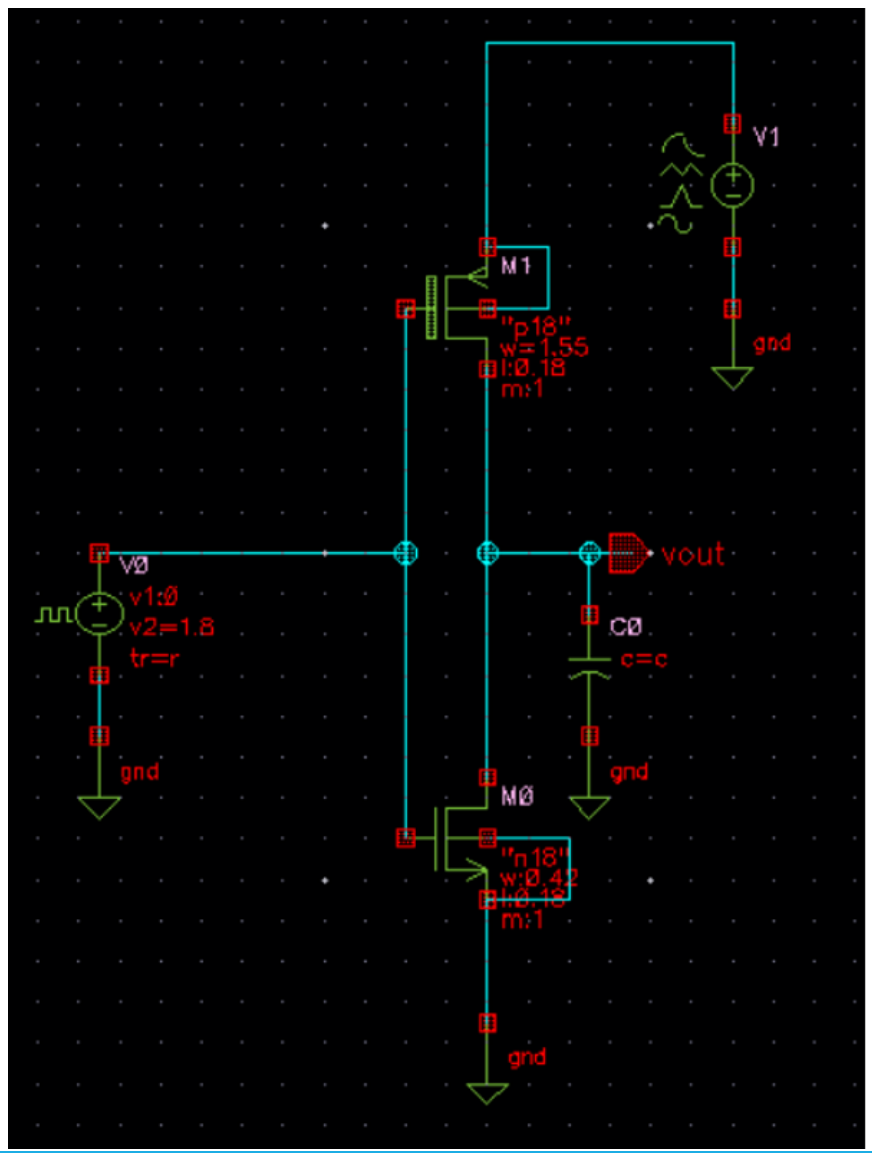
Following Design Parameters were used for inverter design

Gate Length(Same for PMOS NMOS)	Gate Width PMOS	Gate Width NMOS	V _{DD}	C _{parasitic(d,g+d,b)}	V _{th} PMOS	V _{th} NMOS
180 nm	1550 nm	420 nm	1.8V	4fF	-502 mV	468mV

(so the C_{Load} is varied from 100fF to 4500 fF ; in order to ignore the parasitic capacitances effects)

$$I_{ON} \text{ (PMOS)} = 374.543 \mu\text{A}$$

$$I_{ON} \text{ (NMOS)} = 243.61 \mu\text{A}$$



$$I_{\text{dnmos}} (V_{\text{GS}} = 0.5 V_{\text{DD}}, V_{\text{DS}} = V_{\text{DD}}) = I_{\text{NL}} = 48.271 \mu\text{A}$$

$$I_{\text{dpmos}} (V_{\text{GS}} = 0.5 V_{\text{DD}}, V_{\text{DS}} = V_{\text{DD}}) = I_{\text{PL}} = 78.67 \mu\text{A}$$

$$I_{\text{dnmos}} (V_{\text{GS}} = V_{\text{DD}}, V_{\text{DS}} = 0.5 V_{\text{DD}}) = I_{\text{NH}} = 233.78 \mu\text{A}$$

$$I_{\text{dpmos}} (V_{\text{GS}} = V_{\text{DD}}, V_{\text{DS}} = 0.5 V_{\text{DD}}) = I_{\text{PH}} = 327.25 \mu\text{A}$$

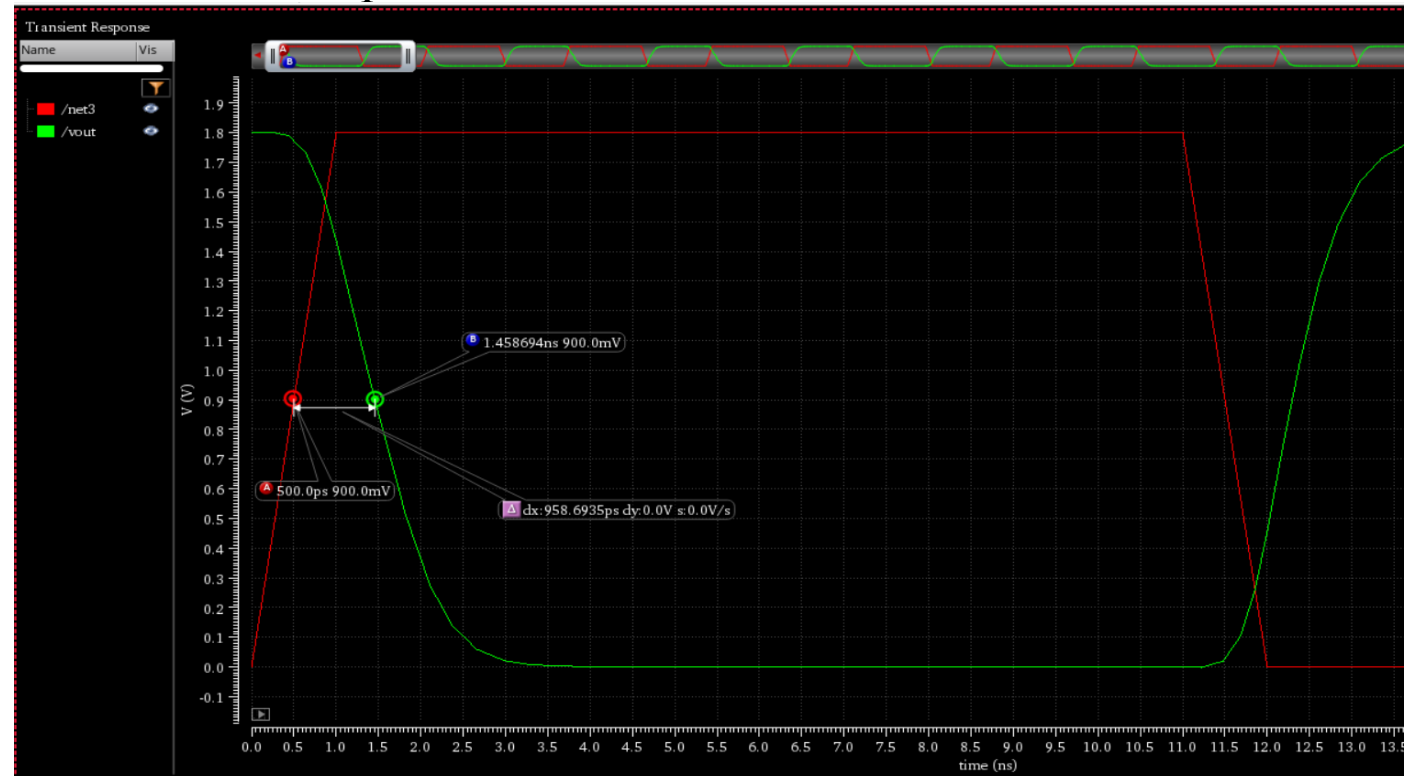
$$I_{\text{dnmos}} (V_{\text{GS}} = 0.75 V_{\text{DD}}, V_{\text{DS}} = 0.75 V_{\text{DD}}) = I_{\text{NM}} = 138.112 \mu\text{A}$$

$$I_{\text{dpmos}} (V_{\text{GS}} = 0.75 V_{\text{DD}}, V_{\text{DS}} = 0.75 V_{\text{DD}}) = I_{\text{PM}} = 201.743 \mu\text{A}$$

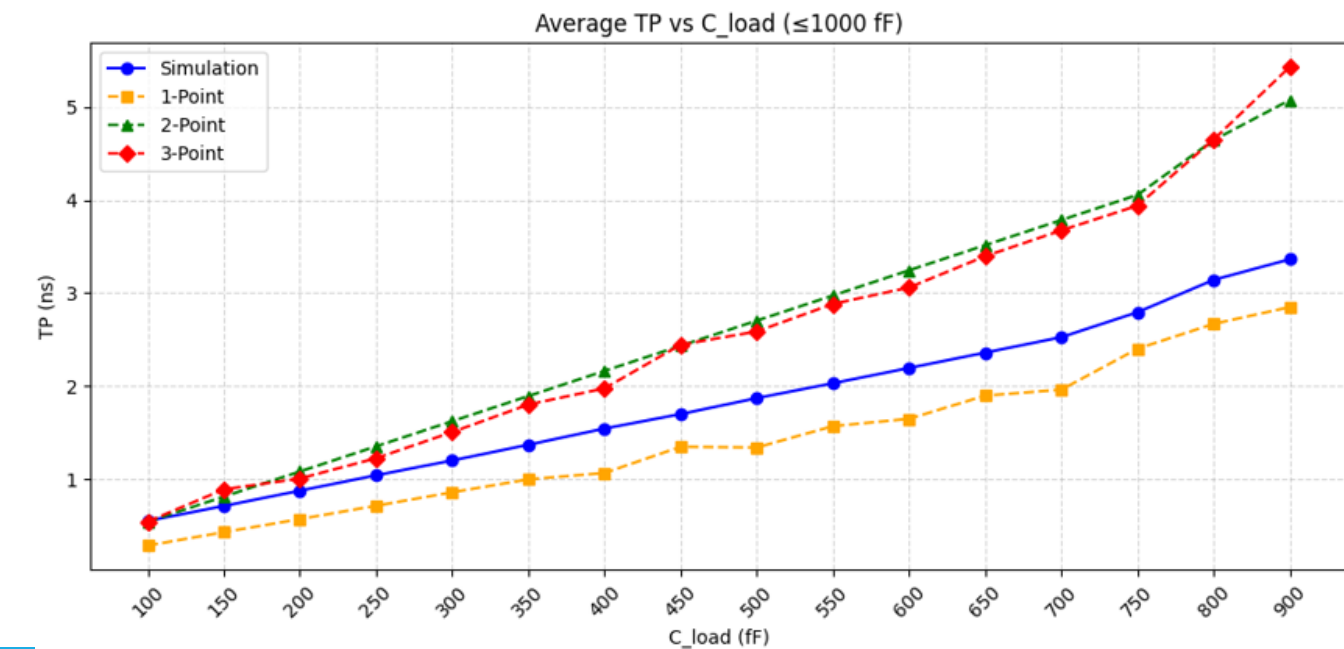
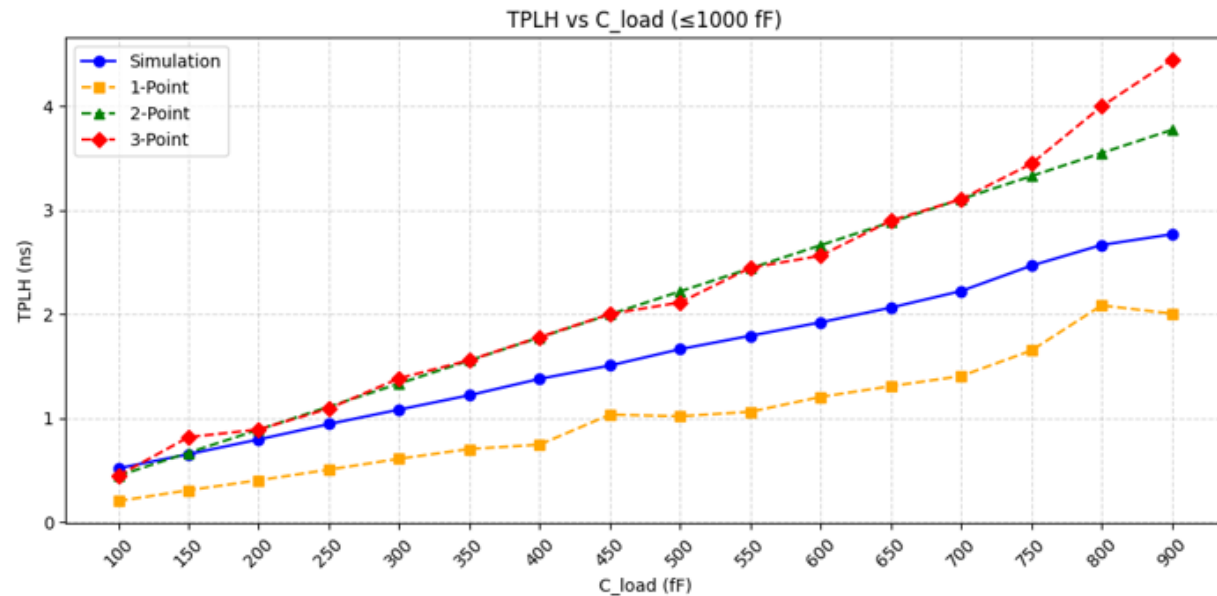
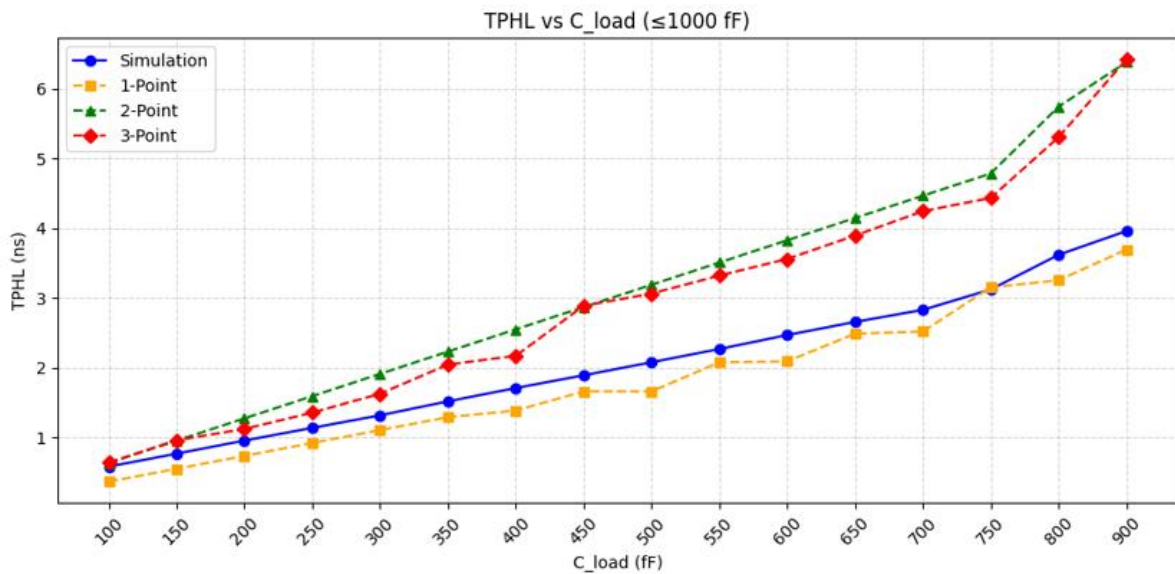
Input Pulse:-

- Time Period = 100ns
- ON Time = 55ns
- Rise time = fall time = 1ns
- Delay time = 0

From these calculations delay from simulations were tabulated and errors were calculated with respect to simulation delay for different load capacitances



C_{load} (fF)	$\tau_{PHL}(ns)(sim)$	$\tau_{PLH}(ns)(sim)$	$\tau_p(ns)(sim)$	Error % onepoint	Error % twopoint	Error % threepoint
100	0.585	0.515	0.55	48.20609455	1.687545455	1.243636364
150	0.7706	0.651	0.7108	39.88467642	14.10773073	14.62296005
200	0.958	0.793	0.8755	34.92484752	23.52221588	24.07995431
250	1.14	0.941	1.0405	31.55538683	29.9179481	30.50456511
300	1.33	1.078	1.204	29.01998007	34.73052326	35.33887043
350	1.52	1.218	1.369	27.17073192	38.24066837	38.86486486
400	1.71093	1.3755	1.543215	26.16285352	40.15376989	40.78660459
450	1.8934	1.5036	1.6985	24.52757374	43.25777156	43.90462173
500	2.0809	1.6612	1.87105	23.87523583	44.49600492	45.14844606
550	2.27	1.79	2.03	22.8194266	46.50008621	47.16157635
600	2.47	1.919	2.1945	22.11442789	47.83827751	48.50580998
650	2.65799	2.0833	2.370645	21.89331933	48.2579741	48.92740161
700	2.831	2.21	2.5205	20.88611942	50.16978774	50.84784765
800	3.2132	2.4641	2.83865	19.71775879	52.38750815	53.0755817
900	3.62	2.663	3.1415	18.38935795	54.90900844	55.60846729
1000	3.96	2.766	3.363	15.29393993	60.78456735	61.51055605
2000	7.735	5.94342	6.83921	16.69608624	58.12308732	58.83705867
3000	11.4781	8.70453	10.091315	15.31337194	60.74768254	61.47350469
4000	15.224	11.5184	13.3712	14.78207491	37.89891708	62.48653823
4500	17.0903	12.8765	14.9834	14.44537555	30.45992565	63.12852891



Worst Case Errors:-

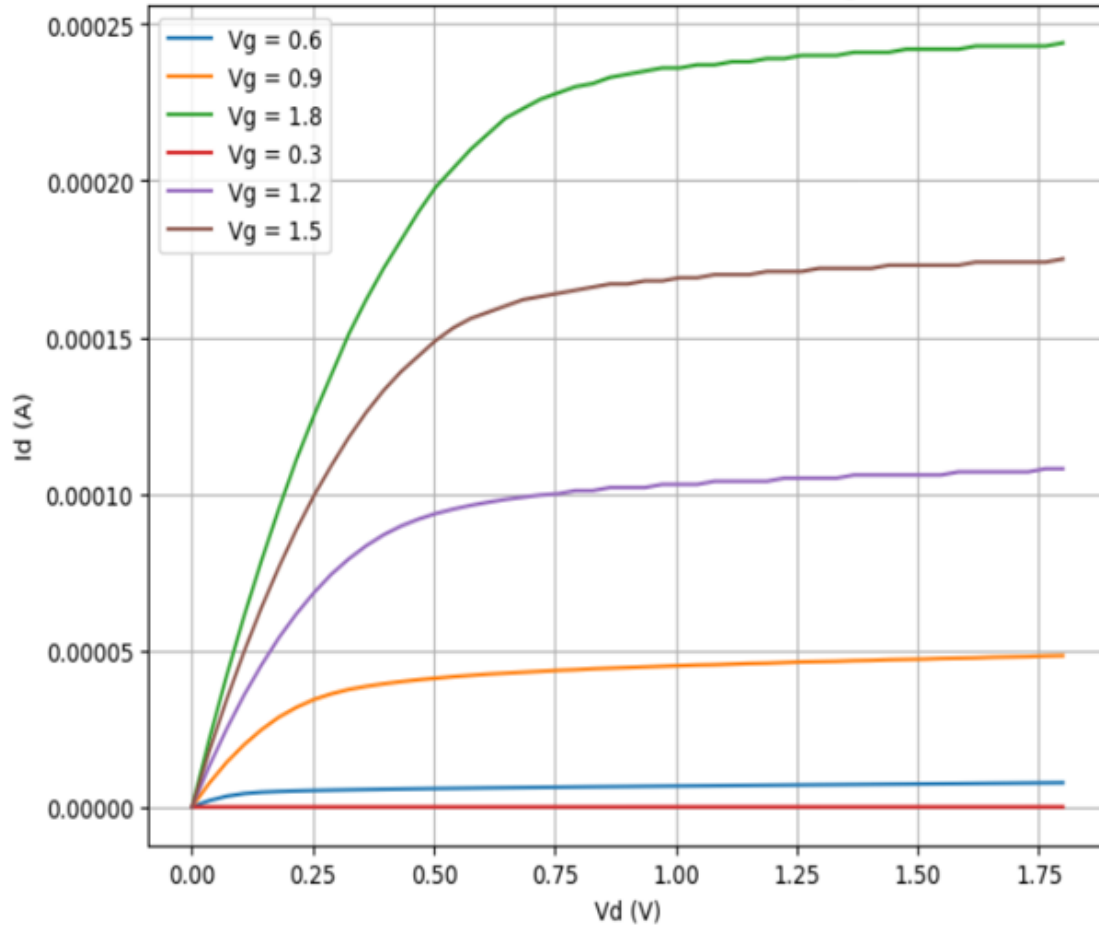
One point model: 48%

Two point model: 60%

Three point model: 61%

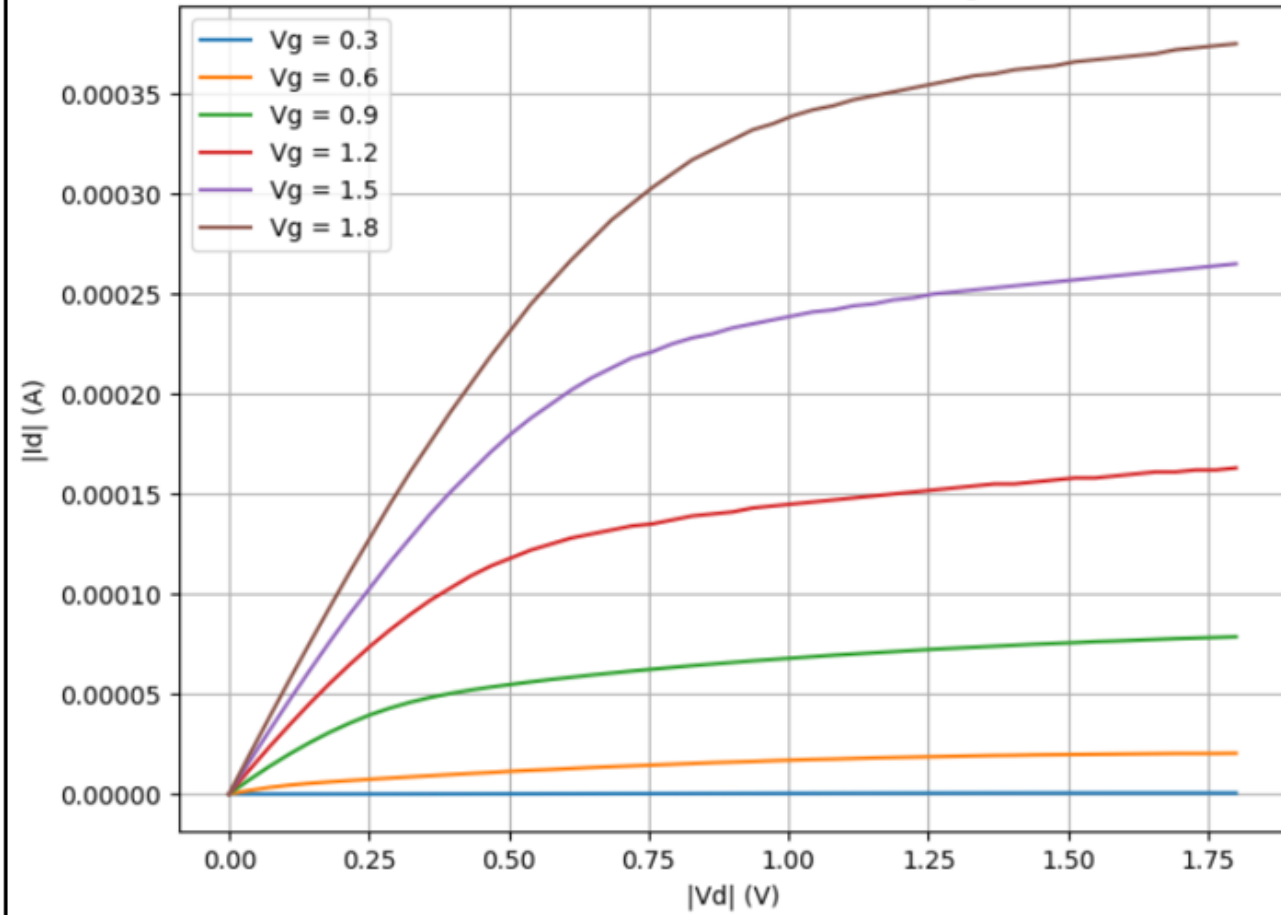
NMOS

Id-Vd Characteristics at Different Vg

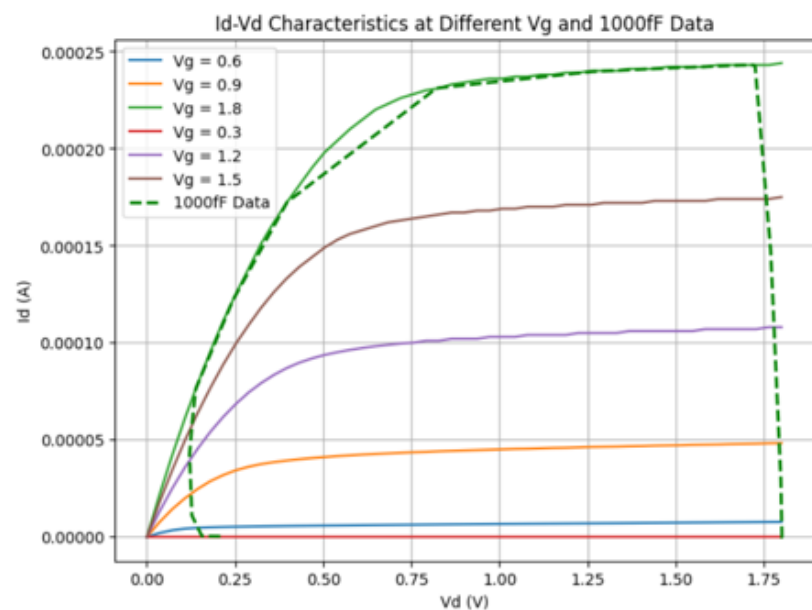
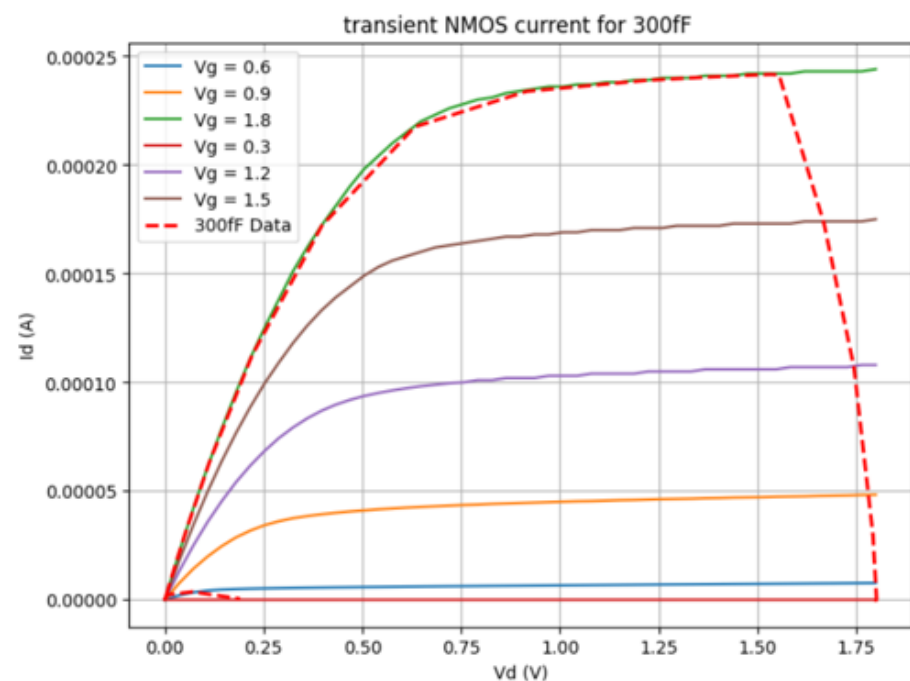
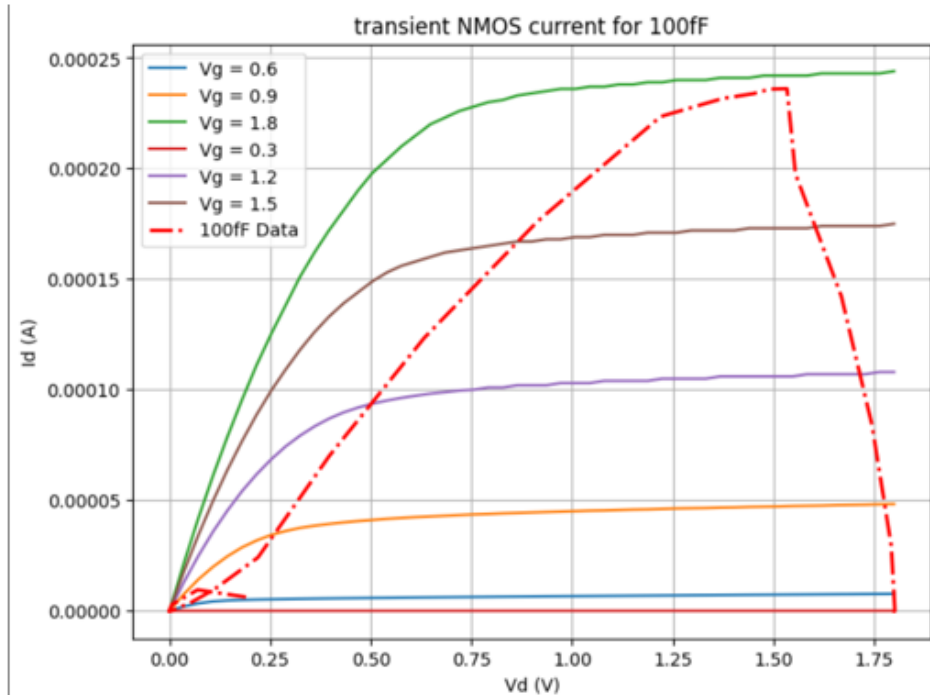


PMOS

Absolute Id vs Absolute Vd for different Vgs (PMOS)



After this we will map the NMOS transient current trajectory of 100fF , 300fF and 1000 fF onto our I_D V_D characteristic curve :-



Peak Currents:-

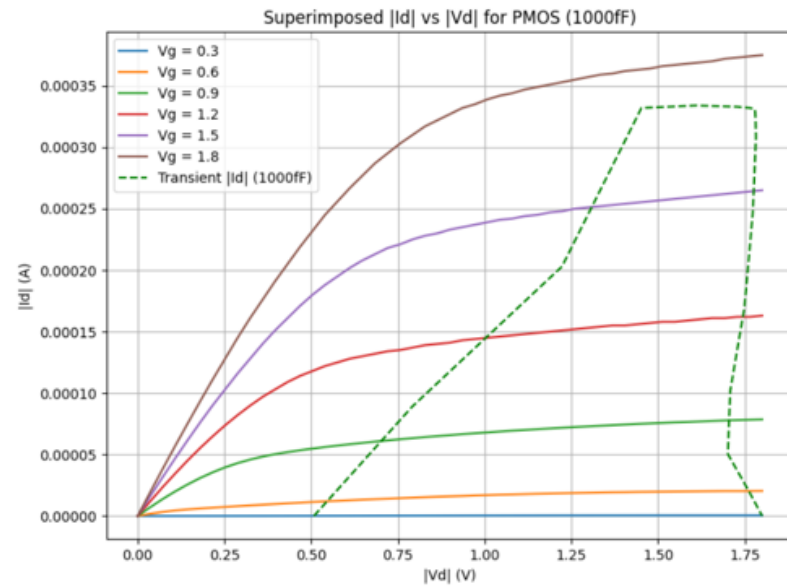
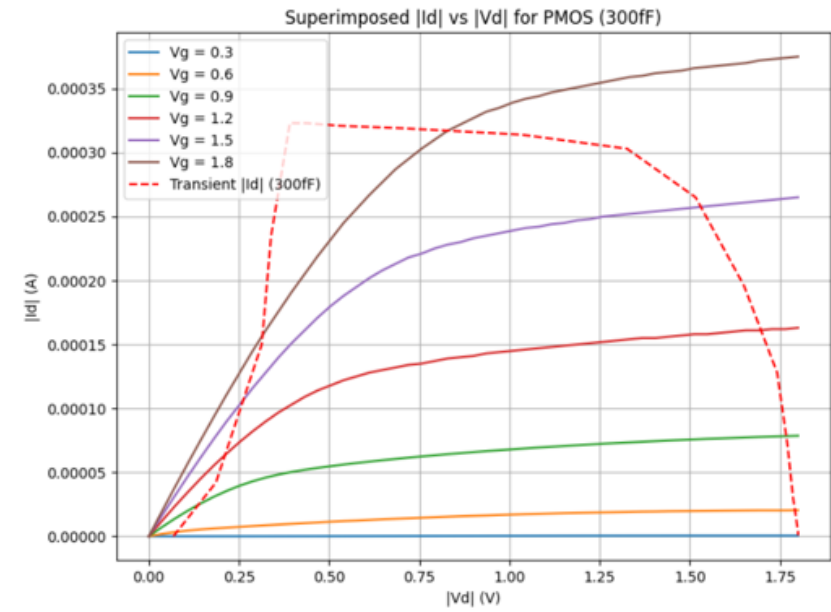
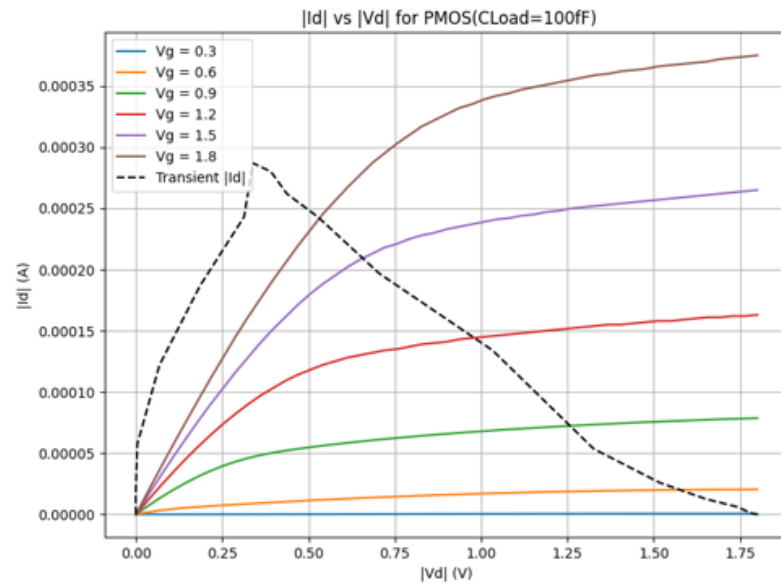
Required Peak :-> $244.10\mu\text{A}$ at $V_{DD} = 1.51\text{ V}$

100fF:-> $236\mu\text{A}$ at $V_{DD} = 1.534\text{ V}$

300fF:-> $242\mu\text{A}$ at $V_{DD} = 1.534\text{ V}$

1000fF:-> $242\mu\text{A}$ at $V_{DD} = 1.489\text{ V}$

Doing The same for PMOS



First Approach :-

τ_{PHL} is different for different capacitances so what if we calculate effective current required from simulation delay results for each and every capacitance and average it and look at what points this current come in NMOS

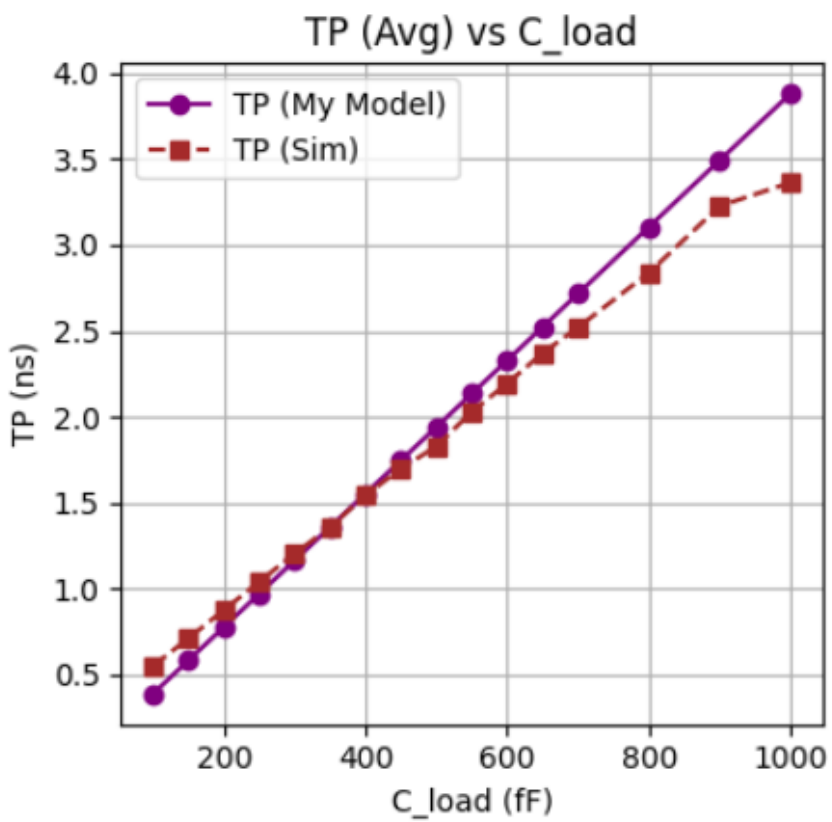
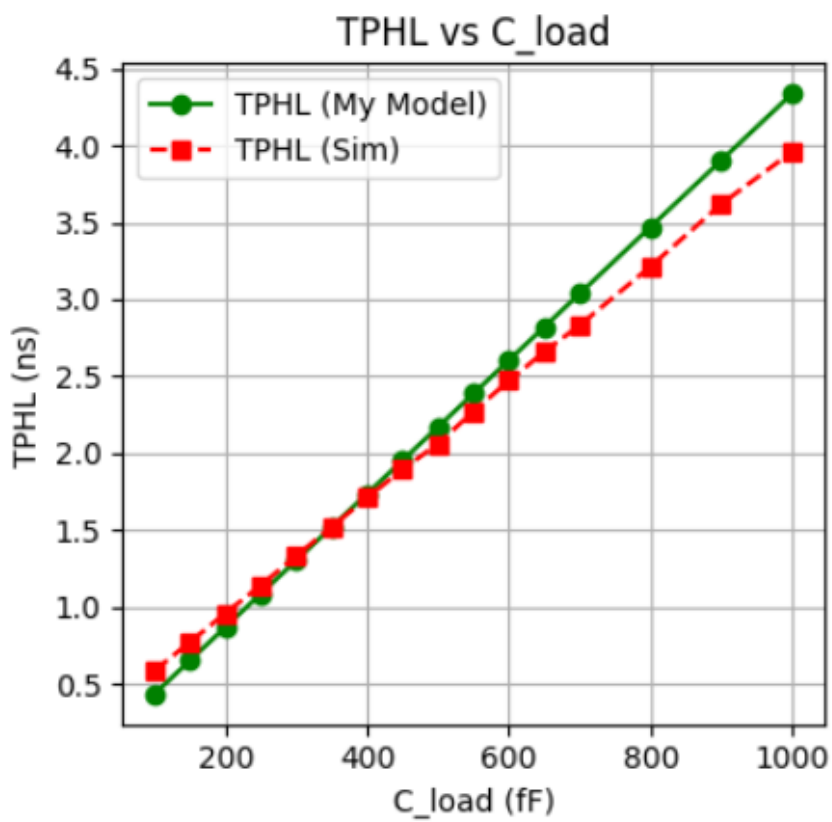
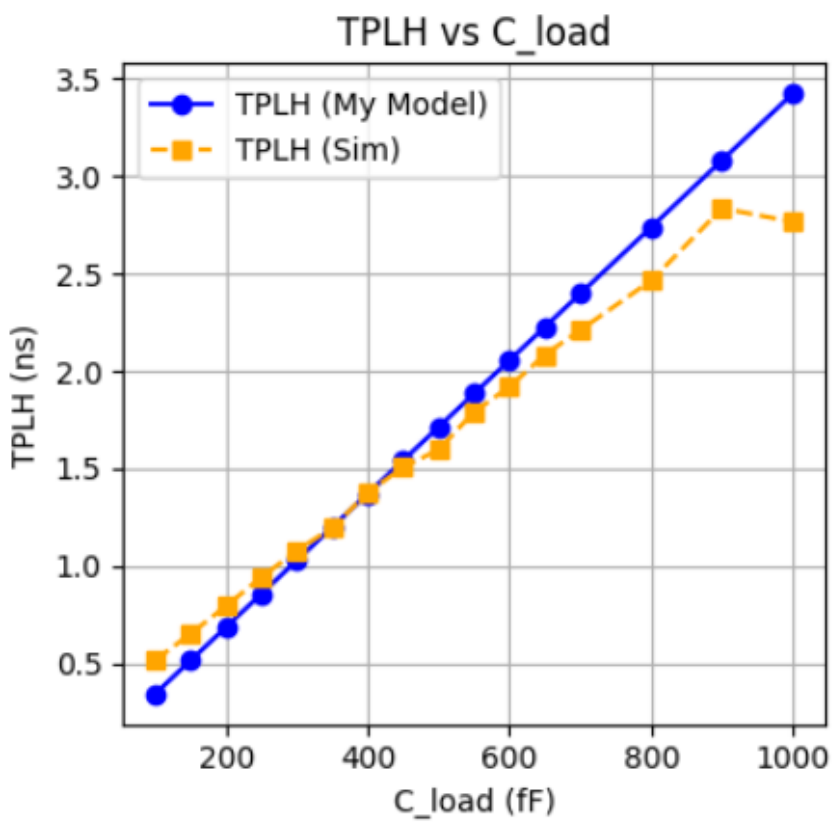
similarly we can do for PMOS and we can formulate an analytical model

For NMOS – > average current is 207.4 μA coming at I_{DS} ($V_{GS} = V_{DD}$, $V_{DS} = 0.3V_{DD}$)

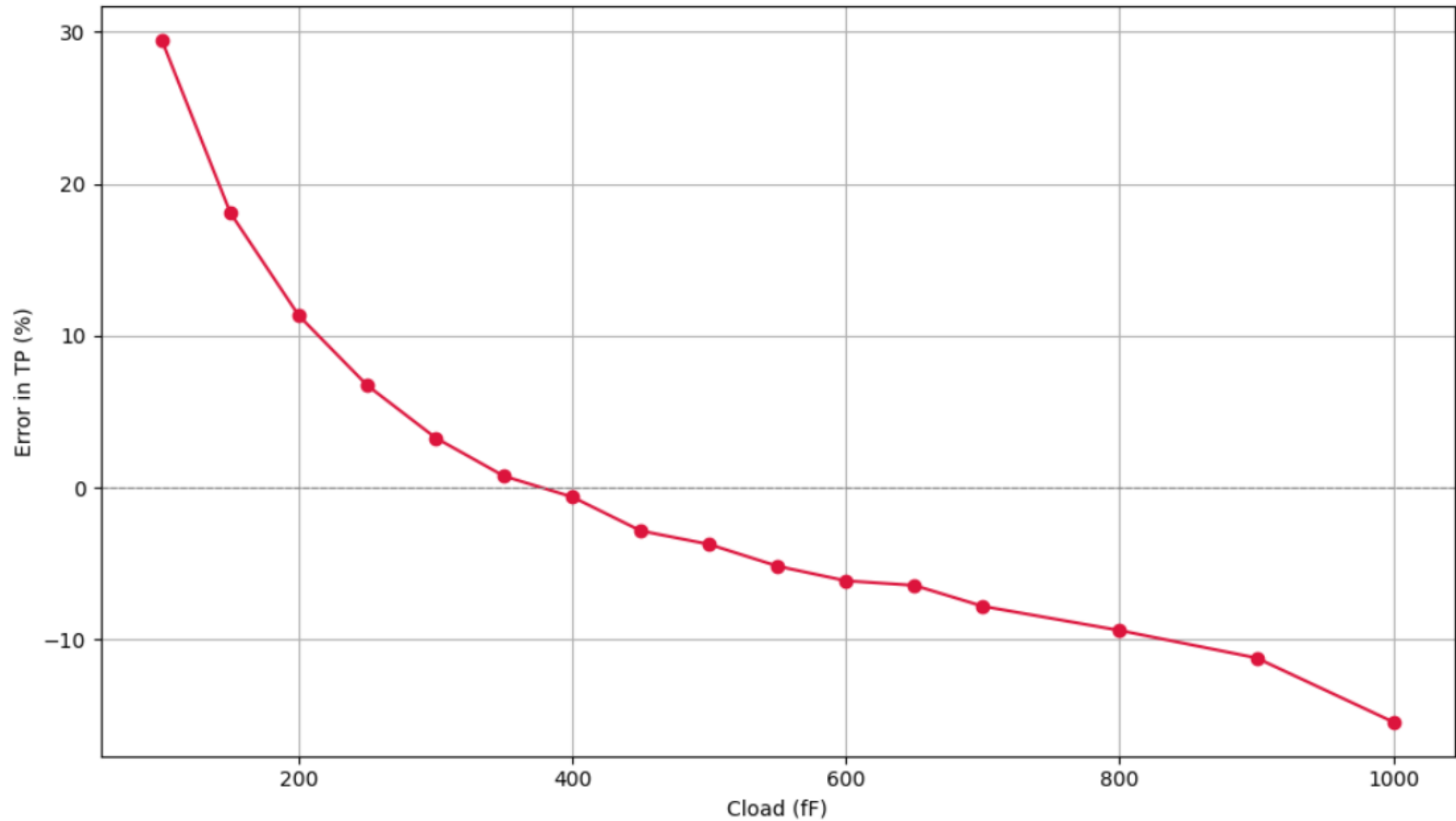
For PMOS -> average current is 262.8014 μA coming at I_{DS} ($V_{GS} = 5V_{DD}/6$ and $V_{DS} = V_{DD}$)

C_{load} (fF)	TPHL(from my model)	TPLH(from my model)	TP	$\tau_p(ns)(sim)$	ERR TP
100	0.433797657	0.342463929	0.388131	0.55	29.43076
150	0.650696486	0.513695894	0.582196	0.7105	18.09283
200	0.867595315	0.684927858	0.776262	0.8755	11.33506
250	1.084494144	0.856159823	0.970327	1.04	6.744163
300	1.301392972	1.027391787	1.164392	1.204	3.289669
350	1.518291801	1.198623752	1.358458	1.369	0.770067
400	1.73519063	1.369855716	1.552523	1.54	-0.60317
450	1.952089459	1.541087681	1.746589	1.6985	-2.83124
500	2.168988287	1.712319645	1.940654	1.871	-3.72005
550	2.385887116	1.88355161	2.134719	2.03	-5.15859
600	2.602785945	2.054783574	2.328785	2.19	-6.11915
650	2.819684774	2.226015539	2.52285	2.365	-6.42041
700	3.036583602	2.397247503	2.716916	2.5205	-7.79272
800	3.47038126	2.739711432	3.105046	2.838	-9.38461
900	3.904178917	3.082175361	3.493177	3.1415	-11.1946
1000	4.337976575	3.42463929	3.881308	3.36	-15.4121

Results From First Approach



Error in TP vs. Cload



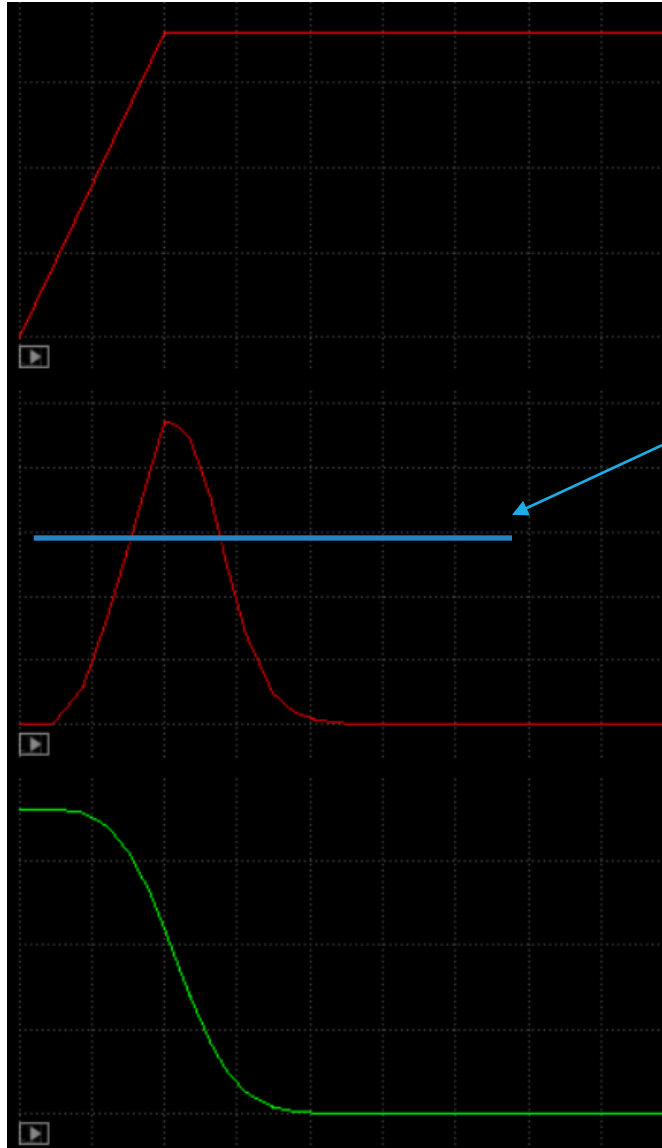
Second Approach :-

Find the average transient current during charging and discharging and fit it to get $I_{\text{effective}}$

Input Pulse
Going Low to
High =>

NMOS
transient
current =>

Output
Voltage Going
High to Low =>



Average Value of Current

(similarly we can do for PMOS)

C_LOAD	T_PHL(sim)(ns)	Average Transient Current NMOS (μA)	TPHL (using 214.14 μA as I_{eff})	Error in TPHL
100	0.585	93.3	0.420285794	28.15627447
150	0.7706	93.3	0.630428692	18.1898921
200	0.958	104.37	0.840571589	12.25766298
250	1.14	113.57	1.050714486	7.832062645
300	1.33	120.64	1.260857383	5.198693006
350	1.52	104.09	1.47100028	3.223665777
400	1.71093	106.6	1.681143177	1.740972608
450	1.8934	101.875	1.891286075	0.111647062
500	2.0809	106.42	2.101428972	-0.986542924
550	2.27	109.2	2.311571869	-1.831359862
600	2.47	108.4	2.521714766	-2.093715224
650	2.65799	108.7	2.731857663	-2.779079801
700	2.831	109.9	2.94200056	-3.92089581
800	3.2132	111.07	3.362286355	-4.639809371
900	3.62	110.476	3.782572149	-4.490943344
1000	3.96	111.35	4.202857943	-6.132776349
		Average = 107.07		

Average of all the $I_{\text{avg,transient}} = 107.07 \mu\text{A}$

To fit with required current (empirically) ; multiply by factor of 2

Therefore ; $I_{\text{eff}} = 214.14 \mu\text{A}$

This $I_{\text{eff}} = I_{\text{DS,NMOS}} (V_{\text{GS}} = V_{\text{DD}}; V_{\text{DS}} = V_{\text{DD}}/3)$

Average of all the $I_{avg,transient} = 107.21 \mu A$

To fit with required current (empirically) ; multiply by factor of 2.5

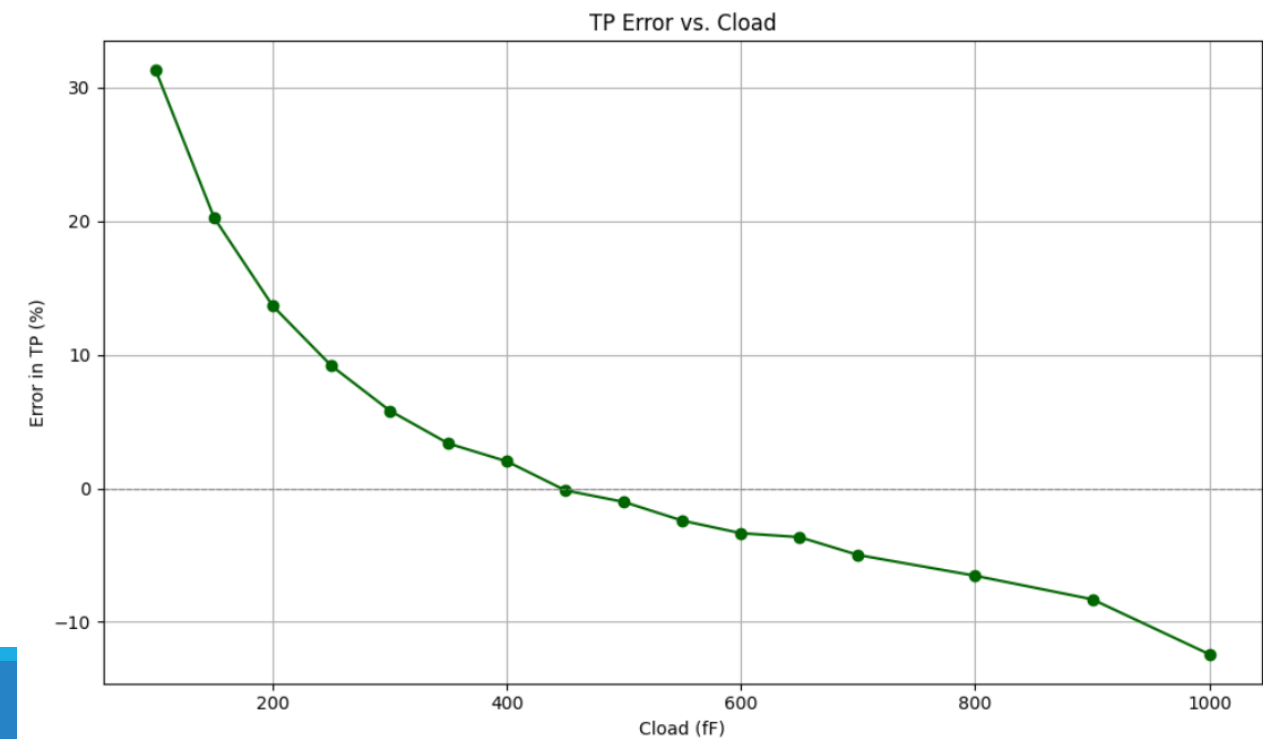
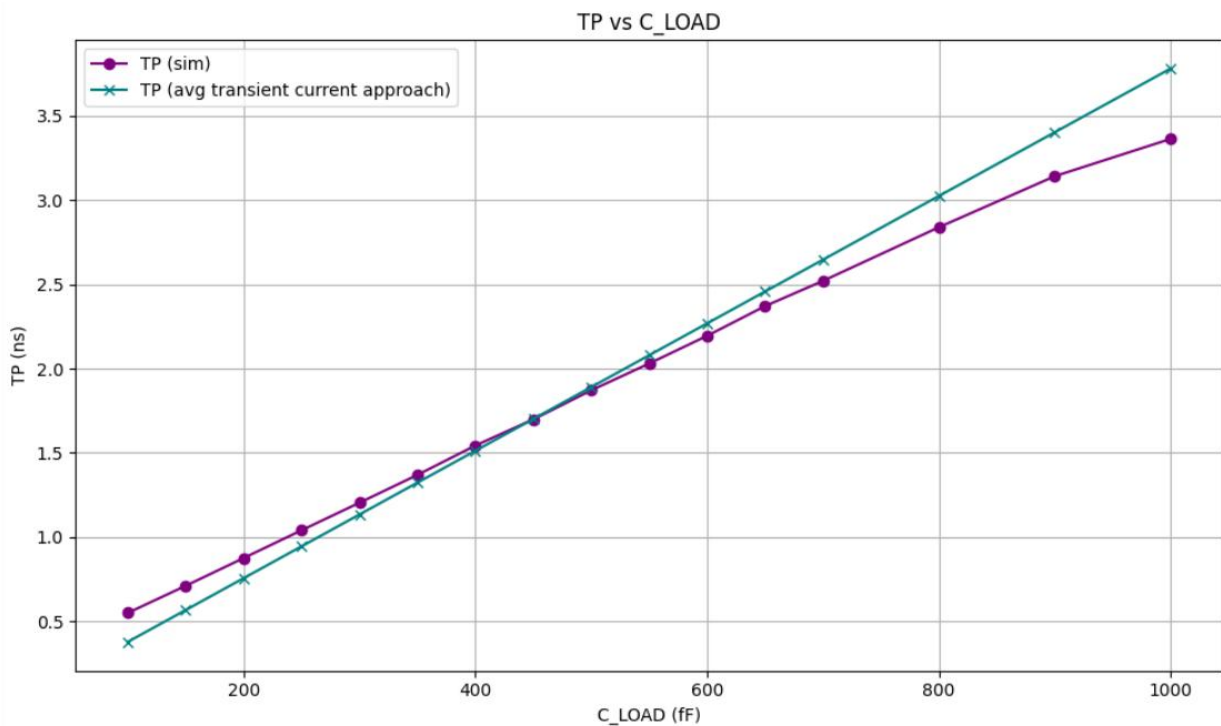
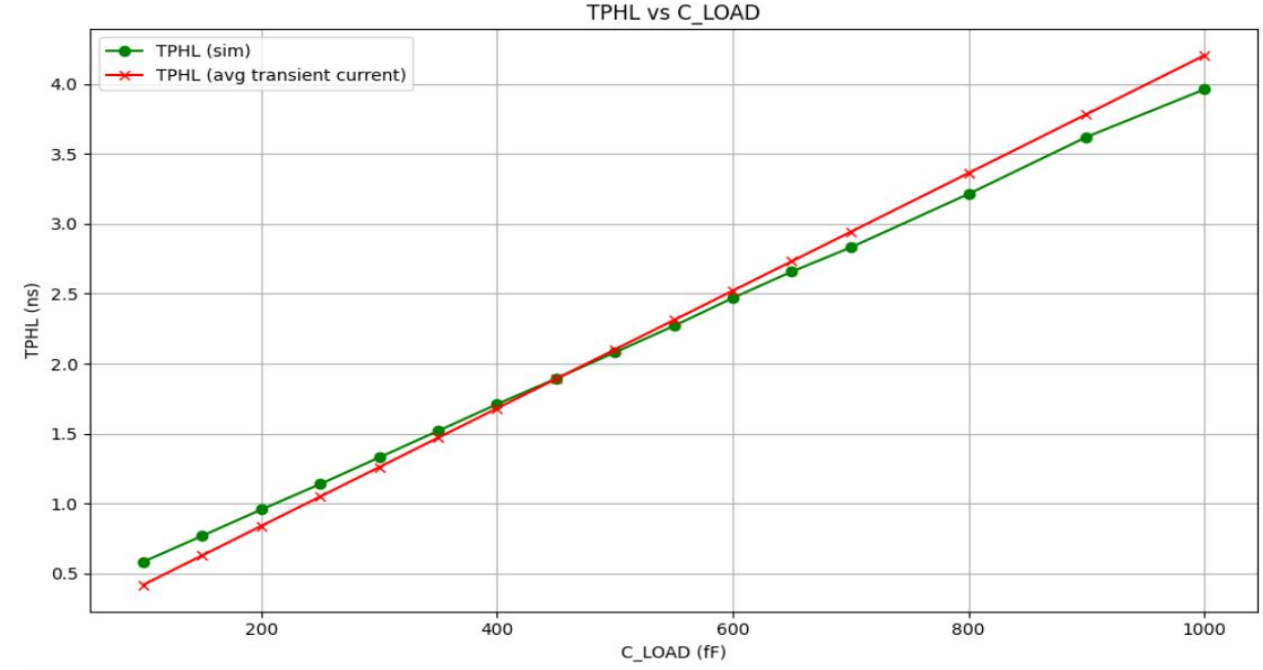
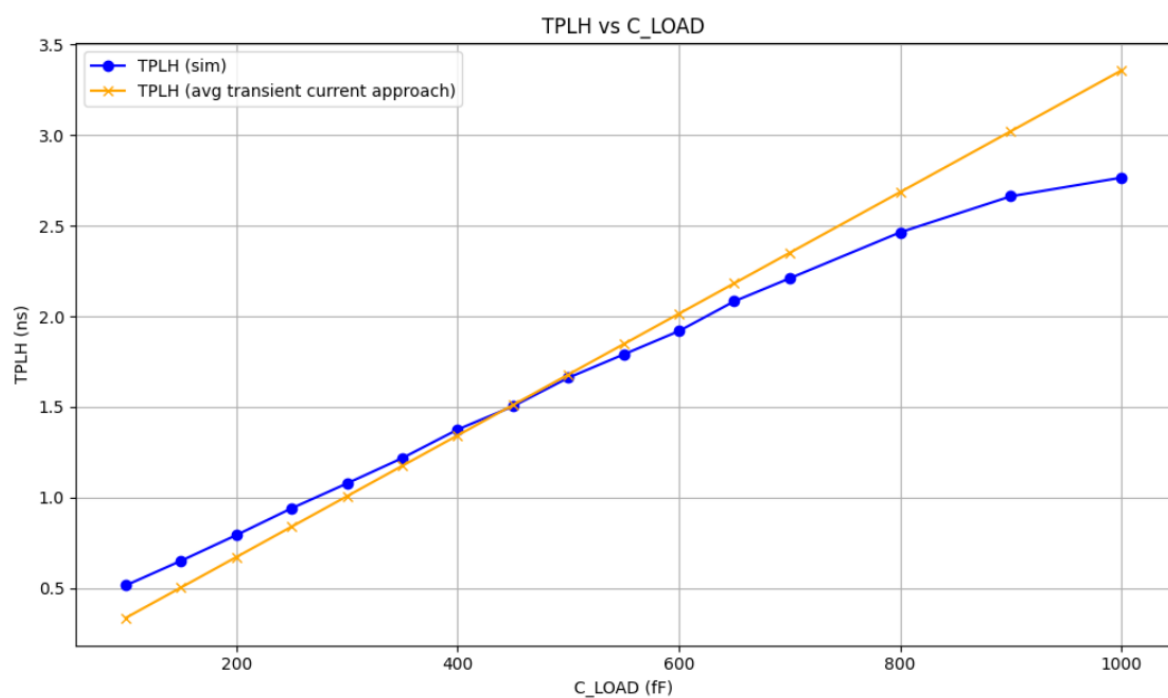
Therefore ; $I_{eff} = 268.025 \mu A$

C_LOAD	T_PLH(sim)(ns)	Error TPLH
100	0.515	-34.79814138
150	0.651	-22.62913091
200	0.793	-15.31158339
250	0.941	-10.78916794
300	1.078	-6.552067198
350	1.218	-3.508743716
400	1.3755	-2.35126954
450	1.5036	0.495681924
500	1.6612	1.068375833
550	1.79	3.175566775
600	1.919	4.98892294
650	2.0833	4.768022711
700	2.21	6.358687922
800	2.4641	9.01816383
900	2.663	13.48502241
1000	2.766	21.39897754

This $I_{eff} = I_{DS,PMOS} (V_{GS} = V_{DD} ; V_{DS} = V_{DD}/3)$



Same Voltage Points as NMOS



C_LOAD	ERR(TP)
100	31.26588
150	20.22277
200	13.64073
250	9.169226
300	5.804564
350	3.350483
400	2.012958
450	-0.15717
500	-1.02287
550	-2.424
600	-3.35959
650	-3.65301
700	-4.98964
800	-6.54013
900	-8.30301
1000	-12.4108

Conclusion

- From both the approaches ; the worst case errors are same ; Approx.: +30% to -15%
- To form an analytical model effectively ; one can take the average of transient currents for both PMOS and NMOS during low to high and high to low transitions of output voltage respectively
- Multiply the current by a factor in order to get the required delay.
- The resultant current obtained at particular voltage points can be taken as the $I_{\text{effective}}$ in denominator of the delay equation.

Future Work:-

- Validate the same procedure using TCAD simulations
- Develop a Physics Based ML model to train the parameters using the formulated model
- Implement the model for scaling across technologies
- Apply methodology to different logic gates and circuits beyond inverter
- Incorporate parasitics and interconnect effects in delay modeling
- For Short channels equations may change (e.g. Adding DIBL coefficient; applying alpha-power MOS model for current equation)
- Build a compact model library to replace time-consuming simulations

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THANK YOU